Tree-Type Network with Augmentation: Overfitting, Ensembles, Bias-Variance and Loss Landscape

Extension of the earlier tree-of-neural-network-nodes assignment. Dataset: Cats vs Dogs, 32x32 (1,902 train / 476 test).
Code: https://github.com/Parul-gbu/tree-type-nn (folder augmentation_extension/)

1. Setup and innovations

Model. As before: a small CNN trunk (16/32/32 channels, 6,240 parameters) and a tree whose nodes are tiny MLP heads that correct the parent's two error types. **Augmentation** = random horizontal flip, random translation of up to 4 pixels (reflect-pad and crop) and brightness/contrast jitter of +/-20%. No rotation or scale. It is used (i) on the fly while training the trunk, and (ii) to build K augmented views of each training image, on which the tree is trained (all views of an image share its label).

What is new here. (a) *View-level augmentation inside the tree*: the tree is fitted to K views per image, and bootstrap resampling for ensembles is done by image, keeping all views. (b) *Bias/variance without leakage*: for every replicate the trunk and tree are retrained from scratch on a different random subset, so trunk variance is included. (c) *Two ensemble types*: shared trunk with bootstrapped trees, and independent trunks; majority vote of hard labels over odd numbers of trees. (d) *Landscape probes* on node heads: top Hessian eigenvalue, filter-normalised 2D slices, interpolation between independent solutions. (e) A test of the hypothesis that augmentation shifts the double-descent peak to the right in proportion to the number of views.

2. Q1: augmentation, overfitting and double descent

Configuration (3 seeds)	Train acc (orig. images)	Test acc	Nodes
A. plain trunk, tree on clean features	96.0%	75.8 +/- 0.5	21
B. plain trunk, tree on 4 aug views	92.4%	75.1 +/- 0.9	43
C. aug trunk, tree on clean features	93.0%	74.6 +/- 0.7	27
D. aug trunk, tree on 4 aug views	84.1%	77.4 +/- 0.6	41

Result. Only the full combination (D) improves on the baseline: +1.6 points on average, and positive in all three paired seeds (+2.5, +1.3, +1.1). Augmenting only the trunk (C) or only the tree (B) did not help (-1.2, -0.7), so the gain is not additive and, with 3 seeds and a 476-image test set, is weak evidence. D also does **not** overfit: it fits only about 84% of its augmented training set at this tree size. To test genuine interpolation we grew bigger trees (width 128, depth 12; 2 seeds). The tree on clean features interpolates (100% train) but drops to 70.6% test, about 3.7 points below the smaller tree C on the same trunks (74.3% on those seeds). The tree on 4 augmented views reaches 93-95% train and 72.6% test. So forcing interpolation costs accuracy, and augmentation limits how far the tree can memorise.

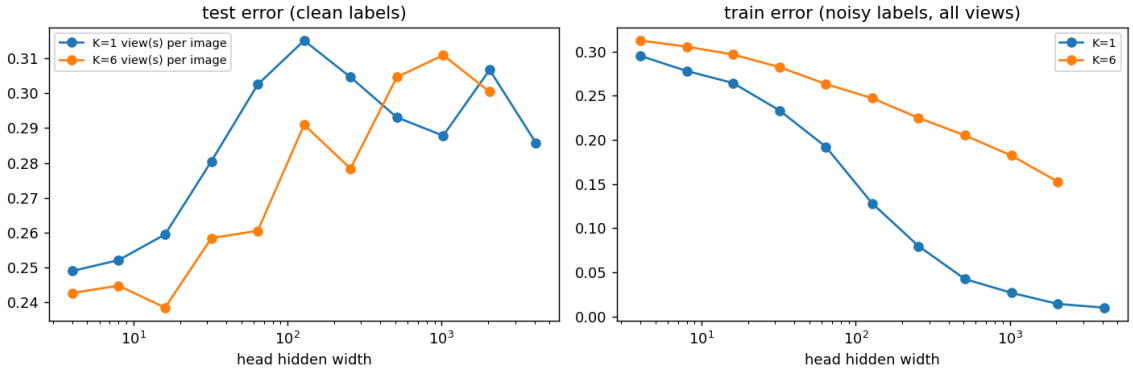


Figure 1. Head-width sweep with 20% label noise on frozen features (K = views per image). Points up to width 1024 use 2 seeds; wider points use 1 seed.

Double descent. Without augmentation, test error rises from 24.9% (width 4) to a peak of 31.5% (width 128), then falls to 28.6-30.7% at widths 1024-4096: a mild second descent of 1-3 points, partly single-seed. With 6 views, training error never gets close to zero (18% at width 1024, vs 2.7% without views) and the peak moves from width 128 to about 1024

(31.1%), then 30.0% at 2048. The shift is roughly 8x, in the same range as the 6x data multiplier, as expected if the peak tracks the number of effective samples. Evidence is suggestive, not conclusive: coarse width grid, one or two seeds. Epoch-wise (Figure 4, right) we saw **no** second descent in 4000 epochs: test error rises from about 23% to 30% (classical overfitting; early stopping is best), and augmentation lowers the late-epoch level only slightly (about 29% vs 30%).

3. Q2: ensembles with majority voting

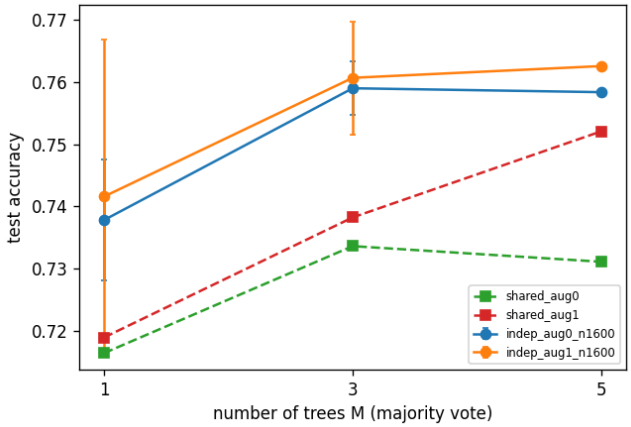


Figure 2. Test accuracy vs number of trees (n = 1600). Independent-trunk curves show mean and spread over all subsets of the 5 trained members.

n	plain: 1 / 3 / 5 trees	aug: 1 / 3 / 5 trees
200	62.6 / 65.4 / 67.4	63.0 / 65.3 / 66.6
800	70.6 / 73.6 / 74.6	69.8 / 73.0 / 74.8
1600	73.8 / 75.9 / 75.8	74.2 / 76.1 / 76.3
1600 shared trunk	71.6 / 73.4 / 73.1	71.9 / 73.8 / 75.2

Ensembling gives about +2 points at n=1600 and up to +4.8 at n=200, saturating by 3 to 5 trees. Independent trunks beat a shared trunk, because the shared trunk's variance is not averaged away (trunk-plus-head predictions alone have variance 0.25-0.29). Augmentation adds little inside the ensemble: differences are within about +/-1 point, except the shared-trunk M=5 case (+2.1, one run).

4. Q3: bias and variance

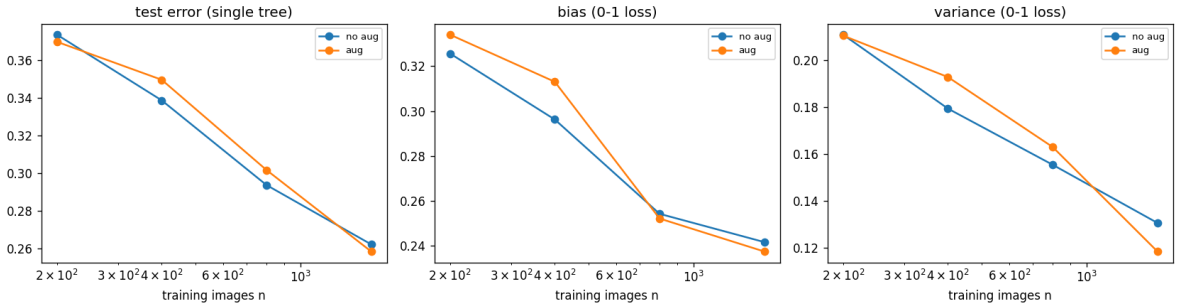


Figure 3. Domingos 0-1 decomposition of a single tree vs training-set size (5 replicates per point, trunk retrained each time).

Error is dominated by bias (0.33 to 0.24 as n goes 200 to 1600) with variance falling 0.21 to 0.13. Because error = bias + (variance on unbiased points) - (variance on biased points), the net variance term (0.047, 0.043, 0.039, 0.021 for n = 200 to 1600, plain trees) is what ensembling can remove. The 5-tree gains in Section 3 equal these numbers almost exactly (+4.8, +4.3, +4.0, +2.0), but note this is partly by construction: with the same 5 replicates, the 5-tree vote *is* the "main prediction" of the decomposition, so its error equals the bias. The independent check is that 3-tree subsets already recover most of the gain. Augmentation did **not** reduce tree variance consistently (lower at n=1600: 0.118 vs 0.131; higher at n=400 and 800). For the end-to-end trunk-plus-head classifier, augmentation lowered test error at all

four sizes (by 4.3, 3.9, 1.5, 2.7 points for n = 1600, 400, 800, 200), but the tree on top removes this difference. Across trunk epochs, bias falls (about 0.47 to 0.23-0.27) and variance rises from near zero to 0.25-0.30, the usual early-to-late pattern, but with 5 replicates the curves are too noisy to resolve finer structure.

5. Q4: loss landscape of node heads

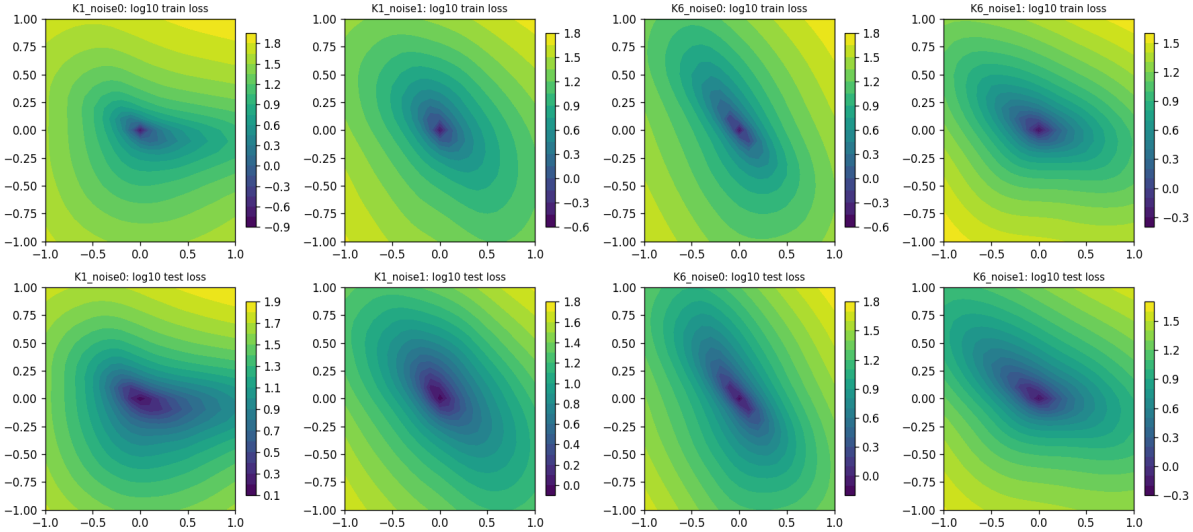


Figure 4 (left part). Filter-normalised 2D slices (log10 loss) around trained width-64 heads. Columns: K=1 clean labels, K=1 noisy labels, K=6 clean, K=6 noisy. Top: train loss, bottom: test loss.

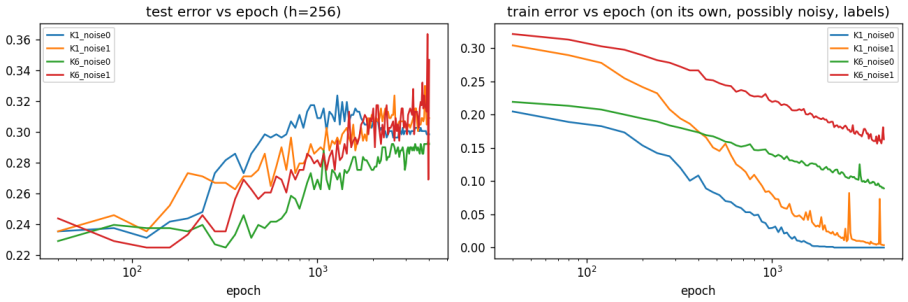


Figure 4 (right part). Test and train error vs epoch for width-256 heads.

Setting (width 64)	Top Hessian eig. (2 seeds)	Test loss	Test err (1 seed)	Interp. barrier (train loss)
K=1, clean labels	41.7 / 41.5	1.32	29.6%	13.2
K=6, clean labels	31.6 / 33.4	0.66	24.8%	2.0
K=1, 20% noise	48.6 / 53.9	0.81	34.5%	3.4
K=6, 20% noise	42.5 / 39.6	0.58	29.0%	2.5

Observations. (1) Minima are narrow and strongly anisotropic: the slices show a thin valley along one diagonal and steep walls in the orthogonal directions, and the train and test slices have the same shape and aligned minima. (2) Augmentation flattens the landscape: top eigenvalue is about 22% lower with clean labels and about 20% lower with noise, edge-of-slice loss is lower (22.8 vs 32.2 clean), and test loss halves with clean labels. (3) The top eigenvalue grows with training (about 3.6 at epoch 40 to 16-47 by epochs 1000-4000 at width 256: progressive sharpening); when the clean data become separable the loss, and its curvature, shrink again (3.4 at epoch 4000, K=1 clean). Power iteration used 25 steps, so values are estimates. (4) Linear paths between two independently trained solutions have a barrier in every setting, as expected for MLPs, and the barrier is much larger for the plain clean-label solutions (13.2 vs 2.0 train loss); this is one pair of solutions and the barrier is not scale-free, so we treat it as an observation rather than evidence of different basins. (5) Flatter minima with lower test loss under augmentation is consistent with the flat-minima account of generalisation, but this is correlational, with a single trunk.

6. Conclusions

(i) Augmentation gives at best a small test gain for the tree (+1.6 points, 3 seeds; the ensemble and bias-variance runs show differences within about +/-1 point), but a clear gain for the trunk-plus-head classifier and for single heads on frozen features (about 5 points, one seed). (ii) Forcing a tree to interpolate hurts; augmentation prevents full interpolation and acts as a regulariser. (iii) We saw a mild, noisy model-wise second descent without augmentation and a right-shifted peak with augmentation; we saw no epoch-wise double descent. (iv) Majority-vote ensembles of independently trained trees recover most of the net-variance term (about +2 points at n=1600), with 3 to 5 trees enough. (v) Augmented heads sit in flatter, lower-test-loss minima.

7. Limitations and future directions

Limitations: one dataset (Cats vs Dogs; CIFAR-10 not rerun here), 32x32 images, 476 test images (about +/-2 points standard error per accuracy), 3 seeds for Q1 and 5 replicates for bias-variance, single trunk for the sweeps and landscape, simple augmentations. Future work: more seeds and a finer width grid around the predicted peaks; sweep trunk width and augmentation strength jointly; stronger augmentation (mixup/cutout) to see whether trees can still interpolate; per-branch augmentation and trunks; CIFAR-10 with the same protocol; sharpness-aware training of node heads; measure the Hessian and barrier for entire trees rather than a single head.